Generalized Additive Model for Council District I (2023 - 2025 Sales)

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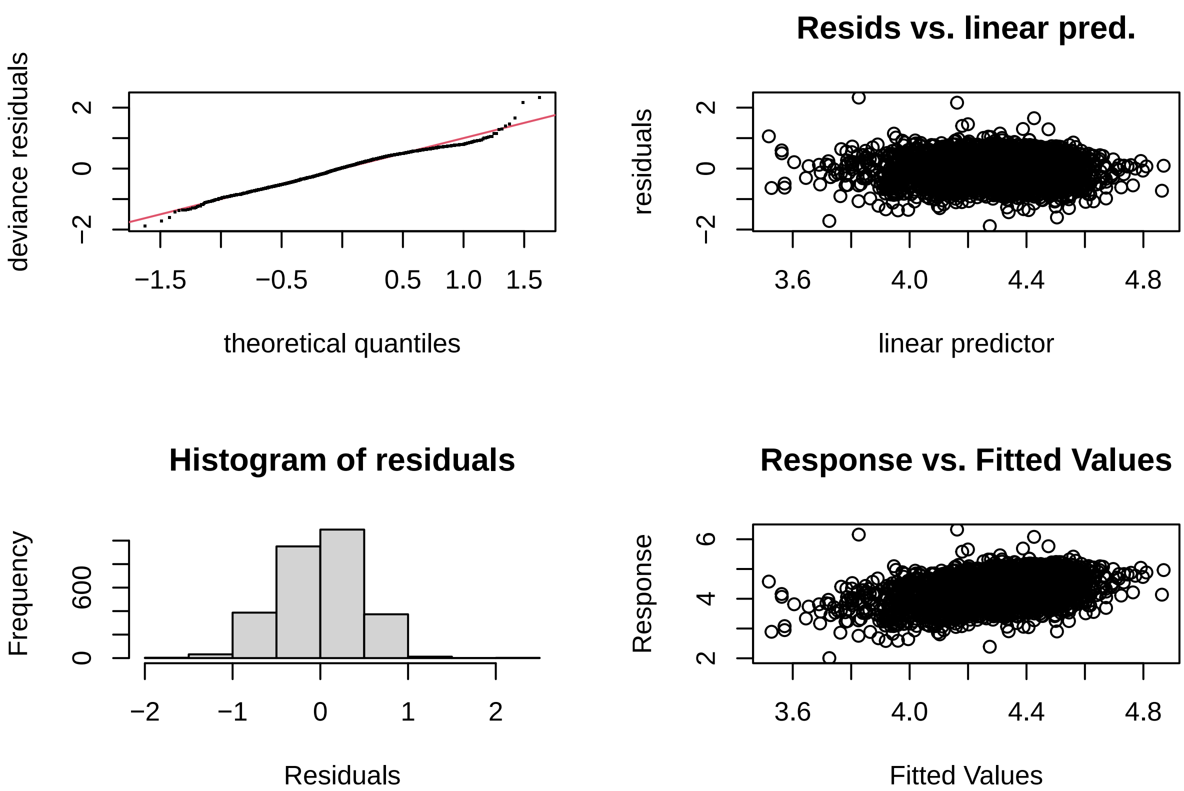
Contents

Model I

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_sppsf ~ arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          4.283953   0.033217 128.967  < 2e-16 ***
## arterial            -0.033711   0.035125  -0.960  0.33727
## single_family1story  0.027912   0.028485   0.980  0.32723
## half_duplex1story   -0.025234   0.121559  -0.208  0.83557
## half_duplex2story   -0.160581   0.190639  -0.842  0.39967
## smallest             0.071248   0.033816   2.107  0.03521 *
## small                0.037922   0.024409   1.554  0.12039
## large               -0.023294   0.027614  -0.844  0.39899
## largest              0.040733   0.055055   0.740  0.45945
## two_plus_baths       0.044208   0.074114   0.596  0.55090
## powder_room          0.008981   0.037346   0.240  0.80997
## good                 0.113675   0.063982   1.777  0.07573 .
## fair                -0.202944   0.023019  -8.816  < 2e-16 ***
## poor                -0.355458   0.062682  -5.671 1.57e-08 ***
```

```
## very_poor_unsound   -0.435543   0.138580  -3.143  0.00169 **
## qabove_avg           0.322763   0.149625   2.157  0.03108 *
## qbelow_avg          -0.028600   0.021790  -1.312  0.18946
## crawl               -0.321365   0.063838  -5.034 5.10e-07 ***
## slab                -0.279781   0.070250  -3.983 6.99e-05 ***
## fireplace            0.091012   0.022179   4.104 4.18e-05 ***
## hot_water            0.002581   0.040314   0.064  0.94895
## elec_baseboard      -0.019831   0.333176  -0.060  0.95254
## wall                 0.089394   0.095922   0.932  0.35144
## central_air          0.067002   0.023769   2.819  0.00485 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                       edf Ref.df      F p-value
## s(sresb_yearbuilt) 3.300      9  6.651  <2e-16 ***
## s(ln_garagespaces) 1.678      3 15.637  <2e-16 ***
## s(months)          4.500      9  7.580  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.133   Deviance explained = 14.3%
## -REML =   1860  Scale est. = 0.20722   n = 2855
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-7.505394e-07,1.733514e-09]
## (score 1860.027 & scale 0.2072173).
## Hessian positive definite, eigenvalue range [0.4626058,1415.506].
```

```
## Model rank =  45 / 45
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                       k'  edf k-index p-value
## s(sresb_yearbuilt) 9.00 3.30    0.91  <2e-16 ***
## s(ln_garagespaces) 3.00 1.68    0.90  <2e-16 ***
## s(months)          9.00 4.50    0.90  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod   prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl> <dbl>
## 1  2855     0.000812   0.000926    0.000706  1.31  46.3 -1.02
```

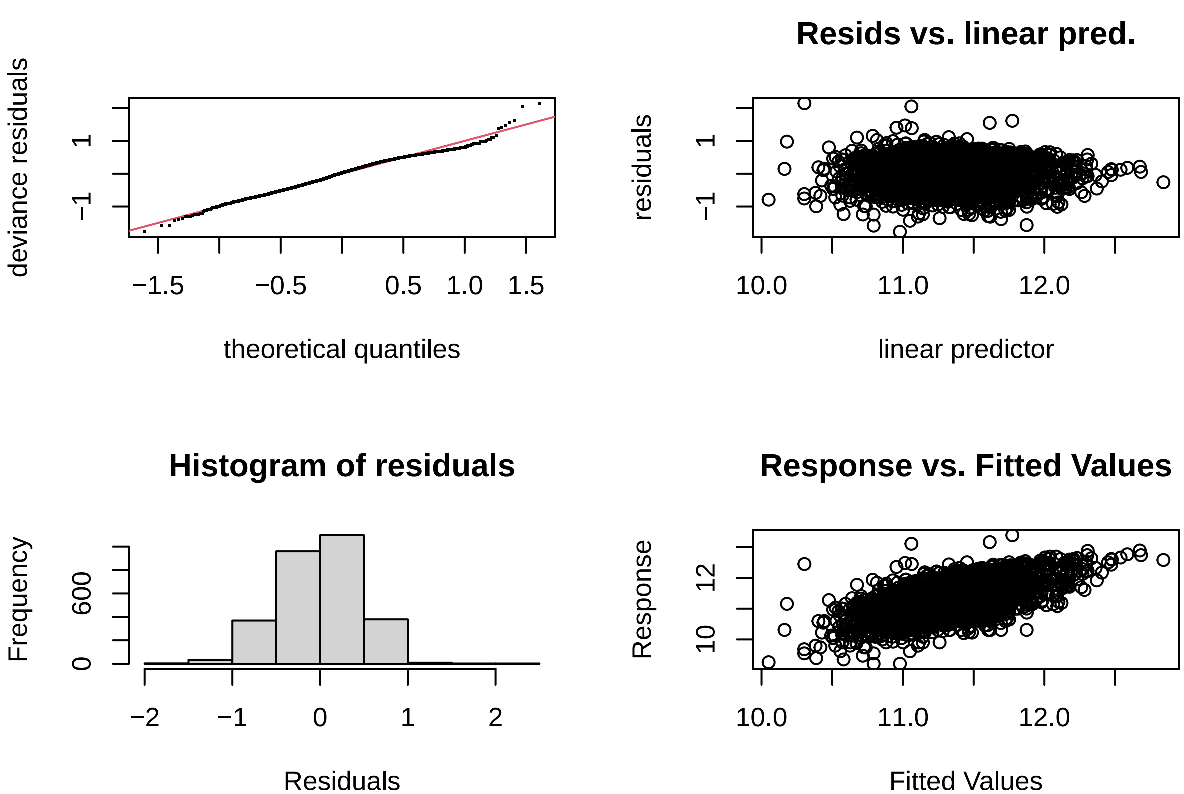
Table 1: District 1 LN_SPPSF NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2855	0.0008866	0.0010059	0.0007665	1.312344	45.77644	-0.000887

Model II

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.327770   0.034432 328.993  < 2e-16 ***
## arterial            -0.004416   0.034924  -0.126 0.899388
## single_family1story  0.008604   0.028359   0.303 0.761612
## half_duplex1story   -0.140227   0.123116  -1.139 0.254807
## half_duplex2story   -0.315176   0.189853  -1.660 0.097003 .
## smallest            -0.128801   0.068945  -1.868 0.061840 .
## small               -0.041046   0.035534  -1.155 0.248139
## large                0.062287   0.039081   1.594 0.111093
## largest              0.168375   0.073718   2.284 0.022442 *
## two_plus_baths       0.183920   0.092653   1.985 0.047237 *
## powder_room          0.060767   0.042573   1.427 0.153588
```

```
## good                     0.144184   0.063397   2.274 0.023023 *
## fair                    -0.193476   0.022806  -8.484  < 2e-16 ***
## poor                    -0.325346   0.062188  -5.232 1.80e-07 ***
## very_poor_unsound       -0.370300   0.137365  -2.696 0.007065 **
## qabove_avg               0.349081   0.150175   2.324 0.020170 *
## qbelow_avg              -0.047144   0.021729  -2.170 0.030115 *
## crawl                   -0.283168   0.063531  -4.457 8.63e-06 ***
## slab                    -0.270503   0.069971  -3.866 0.000113 ***
## fireplace                0.114865   0.022064   5.206 2.07e-07 ***
## hot_water               -0.003831   0.039723  -0.096 0.923183
## elec_baseboard           0.057741   0.329834   0.175 0.861045
## wall                     0.061203   0.095160   0.643 0.520175
## central_air              0.067072   0.023488   2.856 0.004328 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                            edf Ref.df      F  p-value
## s(ln_landratio)          3.297      9  6.740  < 2e-16 ***
## s(ln_base_bldgsize_ratio) 1.464      9  1.256 0.000519 ***
## s(sresb_yearbuilt)       2.814      9  4.983  < 2e-16 ***
## s(ln_garagespaces)       1.592      3 10.503  < 2e-16 ***
## s(months)                4.310      9  8.250  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.357   Deviance explained = 36.5%
## -REML = 1832.6  Scale est. = 0.20256   n = 2855
```




```
##
```

```
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-7.329121e-05,1.301497e-05]
## (score 1832.622 & scale 0.2025618).
## Hessian positive definite, eigenvalue range [0.184497,1415.507].
## Model rank =  63 / 63
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                             k'  edf k-index p-value
## s(ln_landratio)           9.00 3.30    0.92  <2e-16 ***
## s(ln_base_bldgsize_ratio) 9.00 1.46    0.96    0.03 *
## s(sresb_yearbuilt)        9.00 2.81    0.93  <2e-16 ***
## s(ln_garagespaces)        3.00 1.59    0.91  <2e-16 ***
## s(months)                 9.00 4.31    0.91  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2855        0.897       1.02       0.837  1.22  41.1 -0.698
```

Table 2: District 1 LN_PRICE (No ECFs) NLM I Assessing Coefficients

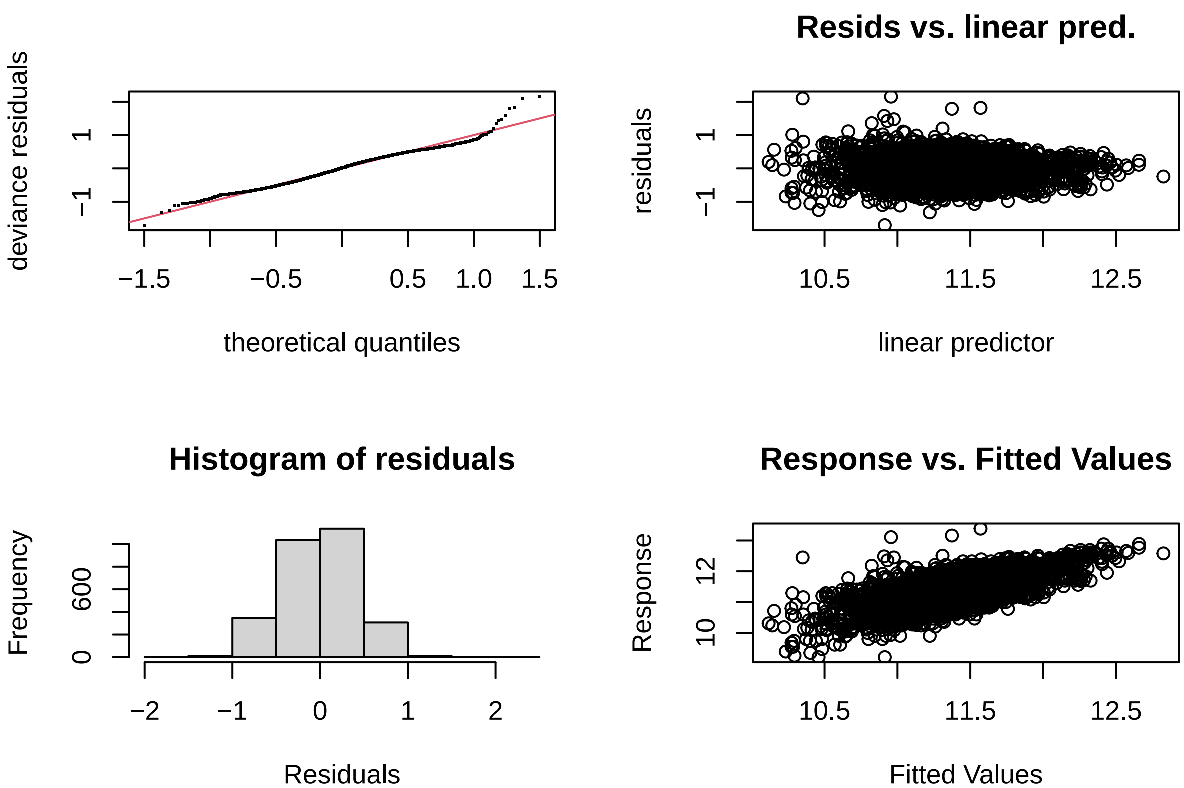
n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2855	0.9747969	1.107624	0.9090016	1.218507	40.72104	-0.7340059

Model III

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10) + ecf
##
## Parametric coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.49582    0.08414 136.626  < 2e-16 ***
## arterial            -0.01710    0.03322  -0.515 0.606688
## single_family1story  0.02701    0.02673   1.011 0.312304
## half_duplex1story   -0.23216    0.11687  -1.987 0.047075 *
```

```
## half_duplex2story   -0.44210    0.17826  -2.480 0.013193 *
## smallest            -0.09533    0.06613  -1.442 0.149510
## small               -0.02715    0.03403  -0.798 0.425065
## large                0.05277    0.03744   1.410 0.158791
## largest              0.10874    0.07022   1.548 0.121617
## two_plus_baths       0.01984    0.08922   0.222 0.824068
## powder_room          0.02186    0.04046   0.540 0.589078
## good                 0.14979    0.05966   2.511 0.012107 *
## fair                -0.13312    0.02170  -6.134 9.76e-10 ***
## poor                -0.26619    0.05833  -4.563 5.25e-06 ***
## very_poor_unsound   -0.21656    0.12851  -1.685 0.092061 .
## qabove_avg           0.23622    0.14142   1.670 0.094964 .
## qbelow_avg          -0.05558    0.02252  -2.468 0.013632 *
## crawl               -0.18100    0.06261  -2.891 0.003871 **
## slab                -0.19497    0.06767  -2.881 0.003993 **
## fireplace            0.06118    0.02209   2.769 0.005664 **
## hot_water           -0.01242    0.03712  -0.335 0.737990
## elec_baseboard      -0.02573    0.30945  -0.083 0.933744
## wall                 0.04394    0.09279   0.474 0.635880
## central_air          0.04284    0.02205   1.943 0.052112 .
## ecf1R102            -0.25308    0.10270  -2.464 0.013785 *
## ecf1R103             0.06746    0.09252   0.729 0.465978
## ecf1R104            -0.02934    0.08260  -0.355 0.722471
## ecf1R105            -0.25886    0.08492  -3.048 0.002325 **
## ecf1R106            -0.21077    0.09392  -2.244 0.024908 *
## ecf1R107            -0.20532    0.12996  -1.580 0.114250
## ecf1R108            -0.32683    0.08585  -3.807 0.000144 ***
## ecf1R109            -0.08130    0.08343  -0.974 0.329952
## ecf1R110            -0.02592    0.09163  -0.283 0.777323
## ecf1R111            -0.16246    0.14985  -1.084 0.278398
## ecf1R112            -0.03850    0.09642  -0.399 0.689686
## ecf1R113            -0.52013    0.09846  -5.283 1.37e-07 ***
## ecf1R114            -0.27726    0.09824  -2.822 0.004802 **
## ecf1R115             0.08168    0.10663   0.766 0.443731
## ecf1R116             0.24976    0.09524   2.623 0.008775 **
## ecf1R117            -0.06422    0.08730  -0.736 0.461978
## ecf1R118            -0.20050    0.09058  -2.213 0.026951 *
## ecf1R119            -0.29637    0.09213  -3.217 0.001311 **
## ecf1R120            -0.03171    0.10142  -0.313 0.754577
## ecf1R121            -0.10026    0.08823  -1.136 0.255934
## ecf1R122            -0.34438    0.08922  -3.860 0.000116 ***
## ecf1R123            -0.20538    0.09529  -2.155 0.031215 *
## ecf1R124            -0.53569    0.11419  -4.691 2.85e-06 ***
## ecf1R125            -0.03773    0.10769  -0.350 0.726075
## ecf1R126             0.28405    0.09327   3.045 0.002346 **
## ecf1R127            -0.34247    0.10076  -3.399 0.000686 ***
## ecf1R128            -0.43808    0.08995  -4.870 1.18e-06 ***
## ecf1R129            -0.09825    0.09332  -1.053 0.292511
## ecf1R130            -0.66603    0.19601  -3.398 0.000688 ***
## ecf1R131            -0.27638    0.12895  -2.143 0.032174 *
## ecf1R132             0.02593    0.10223   0.254 0.799782
## ecf1R133            -0.30630    0.11378  -2.692 0.007146 **
## ecf1R134             0.22849    0.10784   2.119 0.034193 *
## ecf1R135            -0.19036    0.09332  -2.040 0.041460 *
## ecf1R136            -0.23526    0.08777  -2.681 0.007395 **
## ecf1R137            -0.74695    0.10503  -7.112 1.45e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Approximate significance of smooth terms:
##                              edf Ref.df     F  p-value
## s(ln_landratio)            2.876      9 1.136 0.008154 **
## s(ln_base_bldgsize_ratio) 1.608      9 1.735 5.22e-05 ***
## s(sresb_yearbuilt)        1.574      9 1.234 0.000826 ***
## s(ln_garagespaces)        1.425      3 6.736 5.41e-06 ***
## s(months)                 4.369      9 9.589  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.442   Deviance explained = 45.6%
## -REML = 1675.5  Scale est. = 0.17552   n = 2855
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-1.830249e-06,1.337805e-07]
## (score 1675.468 & scale 0.1755221).
## Hessian positive definite, eigenvalue range [0.425148,1397.506].
## Model rank =  99 / 99
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                              k'  edf k-index p-value
## s(ln_landratio)            9.00 2.88    0.98   0.155
## s(ln_base_bldgsize_ratio) 9.00 1.61    1.00   0.440
## s(sresb_yearbuilt)        9.00 1.57    0.99   0.355
## s(ln_garagespaces)        3.00 1.42    1.00   0.530
## s(months)                 9.00 4.37    0.96   0.005 **
```

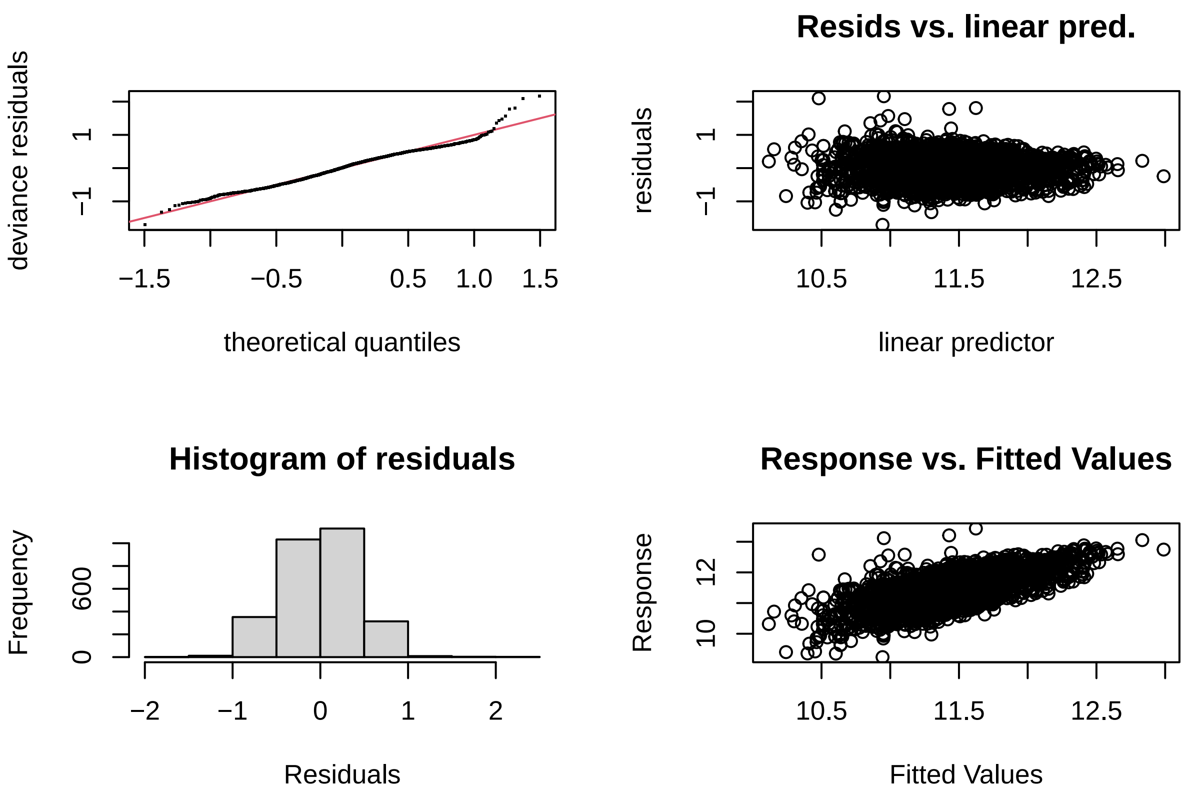
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2855        0.898       1.00       0.850  1.18  37.3 -0.538
```

Model IV

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_tasp ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + ecf + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.57816    0.08351 138.650  < 2e-16 ***
## arterial            -0.01782    0.03318  -0.537 0.591304
## single_family1story  0.02924    0.02457   1.190 0.234074
## half_duplex1story   -0.22619    0.11603  -1.949 0.051341 .
## half_duplex2story   -0.43822    0.17801  -2.462 0.013884 *
## smallest            -0.09529    0.06609  -1.442 0.149455
## small               -0.02689    0.03401  -0.791 0.429180
## large                0.05229    0.03739   1.398 0.162099
## largest              0.10912    0.07018   1.555 0.120063
## two_plus_baths       0.01992    0.08914   0.224 0.823159
## powder_room          0.02199    0.04042   0.544 0.586421
## good                 0.15095    0.05959   2.533 0.011358 *
## fair                -0.13264    0.02164  -6.128 1.02e-09 ***
## poor                -0.26282    0.05805  -4.528 6.22e-06 ***
## very_poor_unsound   -0.21649    0.12817  -1.689 0.091320 .
## qabove_avg           0.24031    0.14116   1.702 0.088792 .
## qbelow_avg          -0.05552    0.02244  -2.474 0.013427 *
## crawl               -0.18190    0.06250  -2.911 0.003637 **
## slab                -0.19480    0.06759  -2.882 0.003982 **
## fireplace            0.06144    0.02207   2.784 0.005406 **
## hot_water           -0.01186    0.03708  -0.320 0.749157
## elec_baseboard      -0.02400    0.30860  -0.078 0.938006
## wall                 0.04583    0.09262   0.495 0.620746
## central_air          0.04328    0.02201   1.967 0.049339 *
## ecf1R102            -0.25078    0.10253  -2.446 0.014508 *
## ecf1R103             0.06868    0.09233   0.744 0.457059
## ecf1R104            -0.02920    0.08249  -0.354 0.723421
## ecf1R105            -0.25895    0.08472  -3.057 0.002260 **
## ecf1R106            -0.21049    0.09382  -2.244 0.024940 *
```

```
## ecf1R107            -0.20549    0.12978  -1.583 0.113447
## ecf1R108            -0.32588    0.08574  -3.801 0.000147 ***
## ecf1R109            -0.08115    0.08333  -0.974 0.330221
## ecf1R110            -0.02620    0.09152  -0.286 0.774710
## ecf1R111            -0.15558    0.14941  -1.041 0.297843
## ecf1R112            -0.03867    0.09630  -0.402 0.688032
## ecf1R113            -0.51834    0.09834  -5.271 1.46e-07 ***
## ecf1R114            -0.27639    0.09812  -2.817 0.004884 **
## ecf1R115             0.08275    0.10642   0.778 0.436866
## ecf1R116             0.24911    0.09501   2.622 0.008790 **
## ecf1R117            -0.06386    0.08720  -0.732 0.464007
## ecf1R118            -0.20053    0.09048  -2.216 0.026744 *
## ecf1R119            -0.29589    0.09200  -3.216 0.001314 **
## ecf1R120            -0.03184    0.10124  -0.314 0.753189
## ecf1R121            -0.09912    0.08809  -1.125 0.260593
## ecf1R122            -0.34347    0.08911  -3.854 0.000119 ***
## ecf1R123            -0.20383    0.09512  -2.143 0.032215 *
## ecf1R124            -0.53330    0.11399  -4.678 3.03e-06 ***
## ecf1R125            -0.03728    0.10758  -0.347 0.728992
## ecf1R126             0.28279    0.09314   3.036 0.002417 **
## ecf1R127            -0.34069    0.10060  -3.387 0.000717 ***
## ecf1R128            -0.43716    0.08985  -4.865 1.21e-06 ***
## ecf1R129            -0.09777    0.09323  -1.049 0.294409
## ecf1R130            -0.67075    0.19574  -3.427 0.000620 ***
## ecf1R131            -0.27621    0.12880  -2.145 0.032076 *
## ecf1R132             0.02552    0.10209   0.250 0.802663
## ecf1R133            -0.30566    0.11365  -2.689 0.007201 **
## ecf1R134             0.22925    0.10772   2.128 0.033413 *
## ecf1R135            -0.19005    0.09321  -2.039 0.041545 *
## ecf1R136            -0.23453    0.08763  -2.676 0.007489 **
## ecf1R137            -0.74571    0.10483  -7.114 1.43e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                               edf Ref.df     F  p-value
## s(ln_landratio)          2.891875      9 1.142 0.008060 **
## s(ln_base_bldgsize_ratio) 1.613430      9 1.755 4.77e-05 ***
## s(sresb_yearbuilt)       1.573122      9 1.235 0.000823 ***
## s(ln_garagespaces)       1.423647      3 6.712 5.67e-06 ***
## s(months)                0.001294      9 0.000 0.651637
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.434   Deviance explained = 44.7%
## -REML = 1665.8  Scale est. = 0.17522   n = 2855
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 9 iterations.
## Gradient range [-0.0004665076,0.0001799468]
## (score 1665.814 & scale 0.1752241).
## Hessian positive definite, eigenvalue range [0.0004662148,1397.503].
## Model rank =  99 / 99
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                                k'     edf k-index p-value
## s(ln_landratio)           9.00000 2.89188    0.98    0.16
## s(ln_base_bldgsize_ratio) 9.00000 1.61343    1.00    0.46
## s(sresb_yearbuilt)        9.00000 1.57312    0.99    0.40
## s(ln_garagespaces)        3.00000 1.42365    1.00    0.55
## s(months)                 9.00000 0.00129    0.96  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2855        0.987       1.09       0.924  1.18  36.6 -0.543
```